Long-Term Effects of Land Reform on Perceived Income Distribution: Evidence from Peru*

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Abstract

This paper analyzes the impact of Peru's agrarian reform on perceptions of income distribution. Using historical data and household surveys, an instrumental variables approach is employed to identify the causal effect of land redistribution on the gap between perceived and actual economic inequality. The results show that in areas with greater reform intensity, the absolute difference between perception and reality is more pronounced. Specifically, a 1-percentage-point increase in redistributed land reduces perceptual bias by an average of 0.127 points, indicating a tendency to underestimate one's actual position in the income distribution. Transmission mechanisms such as changes in income structure, labor formality, and perceived stability are explored, providing relevant evidence for the design of redistributive policies.

Keywords: Land reform, income distribution, biased percecptions
JEL Codes: D13, Q15

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1 Introduction

Studies at the intersection of social and behavioral sciences have demonstrated that individual perceptions play a crucial role in various aspects of social life. The manner in which individuals perceive reality—often diverging from objective reality—shapes their preferences regarding a wide range of public policy issues. A well-documented relationship in this context is that between distributive perceptions and the demand for redistribution: individuals' perceptions of inequality levels in society, as well as their position within the income distribution, influence their preferences for redistribution, which, in turn, affect the intensity of redistributive policies. From this perspective, understanding and quantifying the gap between perception and reality, as well as the factors that explain its existence, persistence, and evolution over time, is essential for designing public policies aimed at reducing preexisting inequalities.

Building on this framework, the present study estimates the effects of a large-scale redistribution of economic resources on perceptions of income distribution. Leveraging the socioeconomic transformations triggered by Peru's agrarian reform in the second half of the twentieth century, this study investigates whether the changes presumably induced by the reform in the income distribution within affected districts have had a lasting impact on individual perceptions. To this end, historical data on the reform are combined with household survey data on perceptions, and an instrumental variables approach is employed to identify the causal effect of the reform on distributive perceptions. The estimated model reveals that in areas where the reform was more intense, the absolute difference between perception and reality is greater. Specifically, after establishing that the bias tends to be negative (that is, individuals tend to underestimate their true position within the distribution), the results indicate that a one-percentage-point increase in redistributed land leads, on average, to a 0.127-point reduction in perceptual bias. Based on these findings, potential transmission mechanisms are analyzed, including changes in the income distribution within reformed areas, labor formality, and perceived stability levels.

The remainder of this study is structured as follows. Section 2 provides background on the agrarian reform and presents descriptive statistics illustrating the existence, magnitude, and direction of bias in perceived income distribution. Section 3 reviews key contributions from the relevant literature. Section 4 describes the data sources and outlines the empirical strategy. Section 5 presents the main estimation results and explores hypotheses regarding the underlying mechanisms. Finally, Section 6 discusses the conclusions.

2 Setting

2.1 Historical context

By the mid-twentieth century, land distribution in Peru was highly concentrated, primarily in large farms and estates, alongside a sector of peasant communities located in the Peruvian highlands. According to the 1961 Agricultural Census, out of a total of 878,667 agricultural holdings, 0.4% comprised more than 500 hectares and accounted for 76% of the total land area. These lands were controlled by landowners belonging to the wealthiest 1% of the population. In contrast, 83% of agricultural holdings comprised less than five hectares and collectively represented only 5.5% of the total available land (Caballero, 1984). It is noteworthy that in 1965, 50% of the economically active population was employed in the agricultural sector (Albertus, 2020).

The agrarian structure of the time was characterized by a system of oppression with colonial roots. The concentration of land ownership perpetuated socioeconomic inequalities and consolidated the economic power of certain elite families. Additionally, precarious labor conditions fueled large-scale mobilizations in the 1950s, taking Peruvian society by surprise. The growing threat posed by subversive groups—such as the Movimiento de Izquierda Revolucionaria (MIR) and the Ejército de Liberación Nacional (ELN)—further contributed to making agrarian reform a central issue in public debate.

Landowners not only wielded economic power but had also accumulated significant political capital at both local and national levels. However, as guerrilla movements encouraged rebellion among marginalized and exploited groups, the first concrete step toward agrarian reform was taken with the enactment of the *Ley de Bases para la Reforma Agraria* in 1963. Nevertheless, its implementation was primarily aimed at dismantling insurgent movements, as it was concentrated in areas influenced by the MIR and ELN.

It was not until the military coup led by General Juan Velasco Alvarado in 1968 that agrarian reform was decisively implemented, coinciding with the abrupt decline of the landowning elite's political influence. Through Decree Law 17716 of 1969, a maximum landholding limit of 150 hectares per parcel was established (with variations depending on location), and landholdings exceeding this threshold were expropriated along with their capital assets. The expropriated land and assets were primarily transferred to agricultural workers, who were organized into cooperatives under the principle of *"the land belongs to those who work it"*. In some cases, land was also allocated to indigenous communities residing in low-productivity areas. In sum, the reform radically transformed land distribution in the country (Albertus et al., 2020; Cleaves and Scurrah, 1980).

2.2 Agrarian Zones

To assess the effects of the Peruvian agrarian reform on individuals' perceptions of their income distribution position, this study leverages agrarian zones—one of the distinctive features of this reform policy. In 1960, prior to Velasco's reform, the *Servicio de Investigación y Promoción Agraria* (SIPA), an agency under the Ministry of Agriculture, established twelve agrarian zones covering the entire country.[1]

Initially, agrarian zones were conceived exclusively for research and development purposes, with technical support from the Organization of American States and the United States Agency for International Development. SIPA delineated these zones based on ecological and social conditions, transportation routes, and market characteristics. All agrarian zones, except for Zone 12, encompassed territory spanning more than one region.

The choice of agrarian zones as administrative units for the reform was not incidental. As noted by Albertus (2020), three key factors influenced this decision: (i) establishing a new organizational structure would have delayed the reform and made it more susceptible to elite resistance; (ii) the limited number of zones facilitated planning, as each zone encompassed expropriated haciendas, small landholders, and indigenous communities, which could be reorganized into large cooperatives and vertically integrated into regional economic centers; and (iii) a smaller number of zones allowed for greater control over the reform and enabled the appointment of trusted leaders for its implementation (Cleaves and Scurrah, 1980).

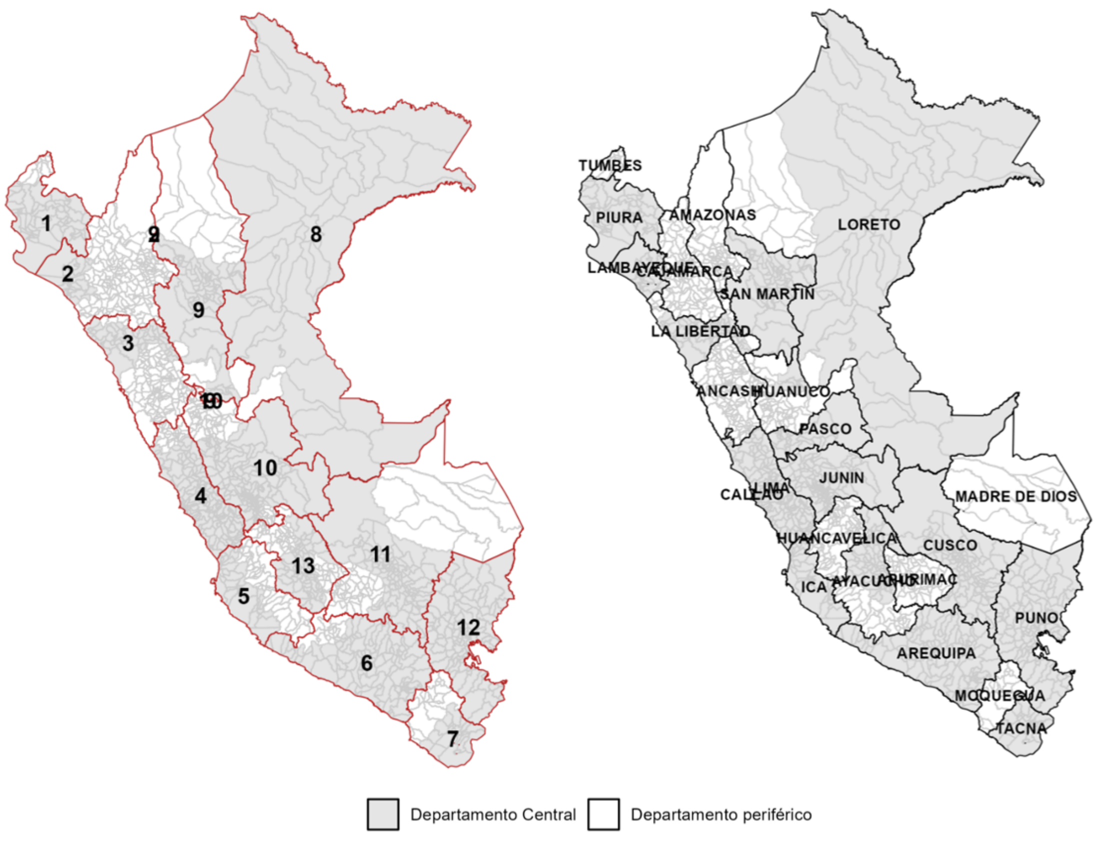
In practice, the use of these administrative units led to differential levels of government atten-

[1] A thirteenth agrarian zone was created in 1974. This detail is relevant as the analysis restricts the sample to agrarian zones whose administrative boundaries remained unchanged.

tion across regions. Since the zones were established prior to the First Agricultural Census, the reform's central offices were logically situated in the most developed cities within each zone. The selection of these cities was not arbitrary, as documented evidence suggests that proximity to the reform's central office influenced the level of exposure to the reform. Echevarria (1978) reports that peripheral areas received minimal attention. Similarly, Albertus (2020) cites official documents from the Ministry of Agriculture indicating that, due to budgetary constraints, resources allocated for training administrative personnel outside the central departments were limited. Furthermore, Condor (2024) finds a negative correlation between the level of land redistribution and distance from the central office.

Figure 1 illustrates the administrative division by agrarian zone and department. The shaded area represents districts belonging to the central departments within an agrarian zone. As observed, districts from the same department may belong to multiple agrarian zones. For clarity, a *central department* is defined as the one hosting the reform's administrative office, meaning that only one department per agrarian zone could be classified as central. This centralization resulted in uneven exposure to the reform, offering an opportunity to examine its long-term effects. In particular, the varying intensity of treatment enables the isolation of common sources of bias, such as confounding factors.

Figure 1: Agrarian Zones and Departmental Boundaries



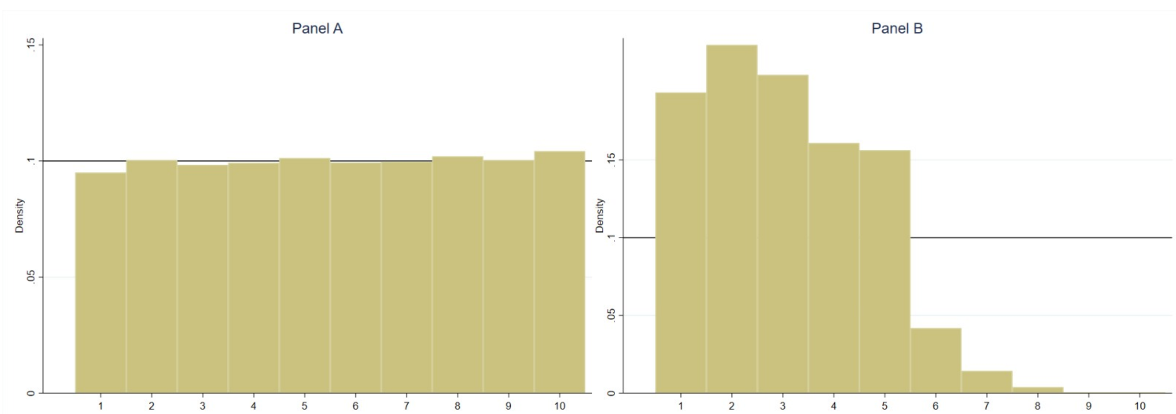
2.3 Perceptions

Given the redistributive nature of the agrarian reform, it is important to analyze whether it led to changes in the perceptions of those living in the affected areas. Although we do not have data for the years in which the reform was implemented, we can turn to current data to characterize these individual perceptions at the national level. We identify potential persistent differences over time between districts based on their level of exposure to the reform.

In particular, we examine the relationship between the deciles of objective income and the deciles of perceived income, as well as the resulting bias, defined as the difference between the two. Additionally, we analyze the distribution of this bias and its evolution in recent years. To this end, we use a dataset constructed from the National Household Survey (ENAHO) for the period 2012-2023. Thus, the variables analyzed reflect the average across all years in the sample.

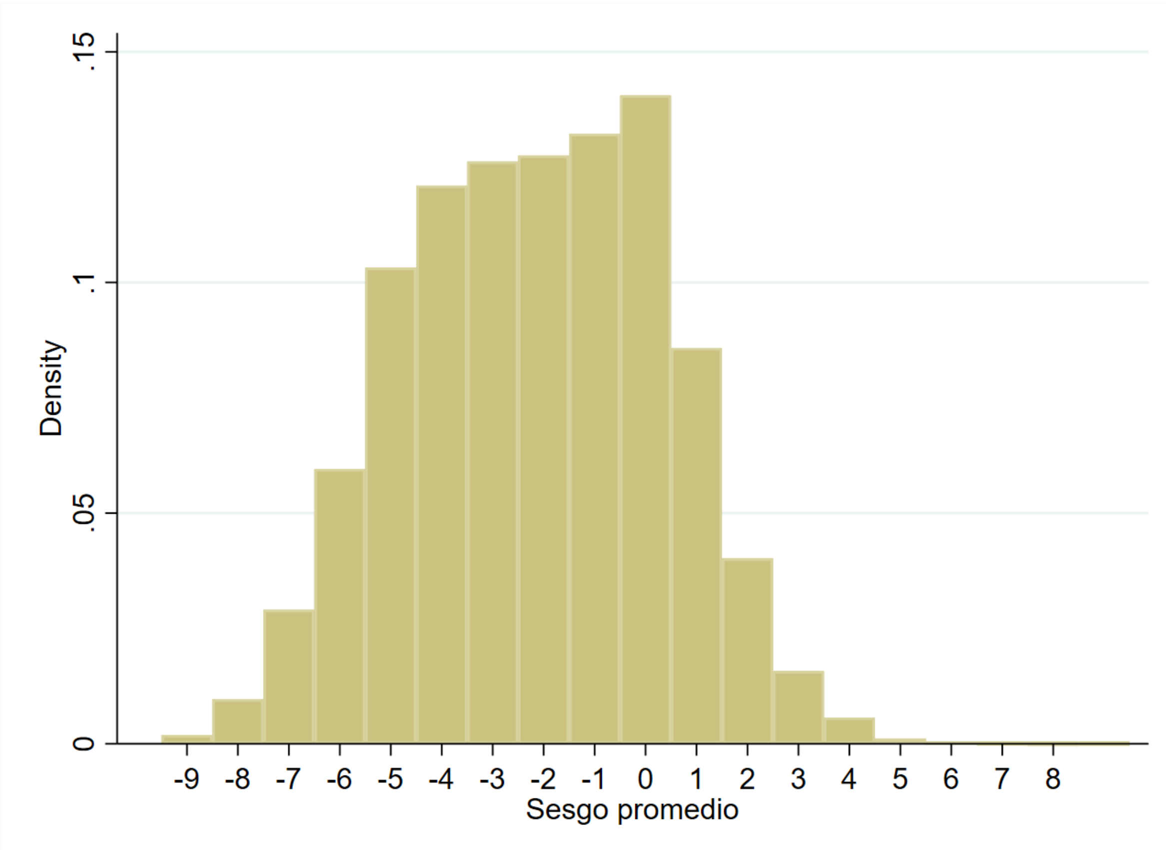
Figure **??** highlights significant differences between the objective income deciles to which individuals belong and the deciles they perceive themselves to be in, confirming the presence of a bias in perceptions. In turn, Figure 3 illustrates that the distribution of the mean bias is concentrated to the left of zero, indicating that, on average, individuals tend to underestimate their position in the income distribution, perceiving themselves to be poorer than they actually are.

Figure 2: Distribution of objective and perceived own-income decile



Source: Author's own elaboration based on ENAHO microdata (2012-2023)

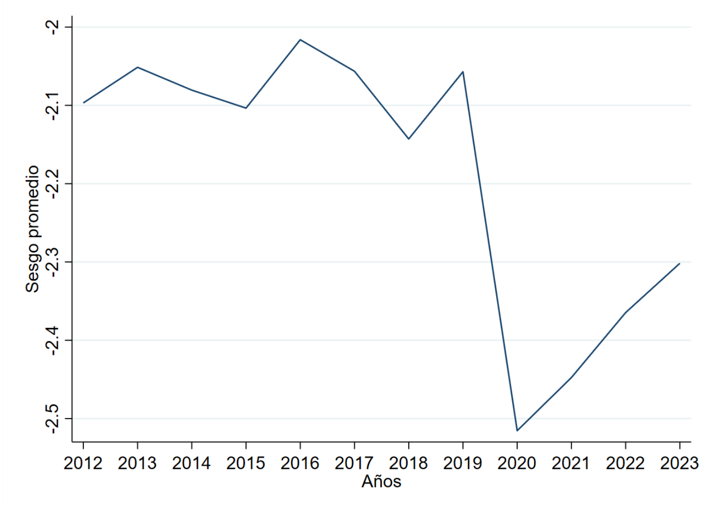
Figure 3: Distribution of the Average Bias



Source: Author's own elaboration based on ENAHO microdata (2012-2023)

Figure 4 shows the evolution of the average bias over the analyzed period. While it remained relatively stable over time—on average, individuals underestimate their position in the income distribution by approximately two deciles—the figure also reveals that individual perceptions are sensitive to short-term factors, such as economic shocks. One example of this is the sharp increase in the average bias in 2020, presumably attributable to the impact of the COVID-19 pandemic.

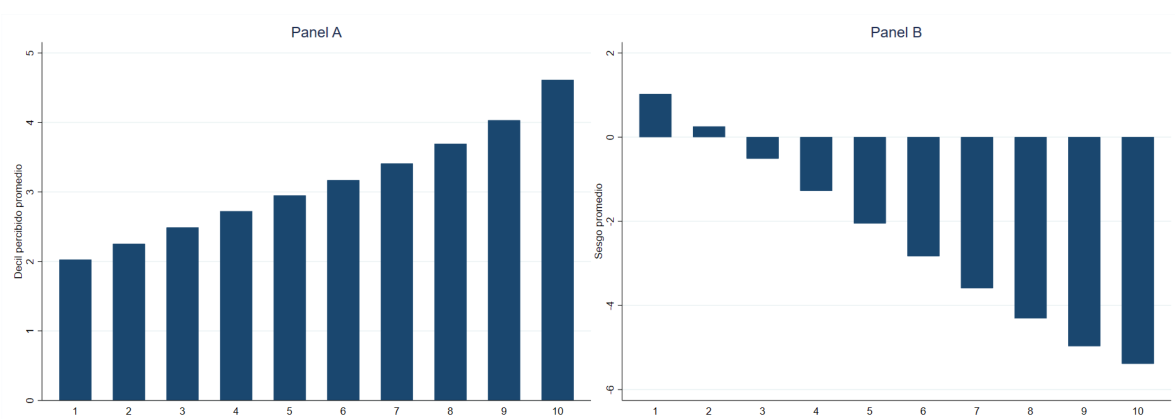
Figure 4: Trends in the Average Bias



Source: Author's own elaboration based on ENAHO microdata (2012-2023)

Figure 5 breaks down the perceived decile and the bias by the objective income decile. In Panel A, it is observed that the bottom 20% of income earners tend to overestimate their position, whereas, starting from the third decile, individuals tend to underestimate their true position. This phenomenon becomes more pronounced as purchasing power increases. In contrast, Panel B shows that the lower deciles exhibit positive biases, and from the third decile onward, the sign of the bias becomes negative. Ultimately, this reveals a positive relationship between income and the perceived decile: the higher one's position in the income distribution, the higher the decile individuals believe they occupy.

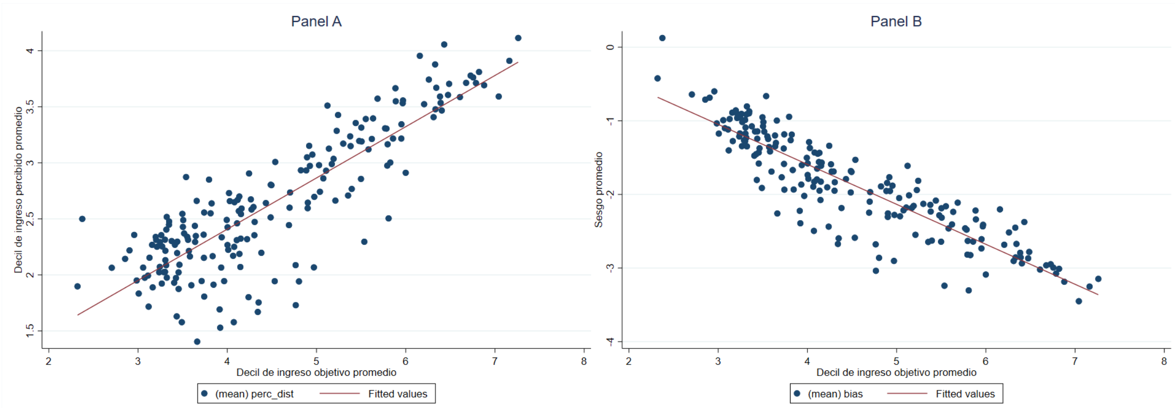
Figure 5: Perceived Income Decile and Average Bias by Objective Income Decile



Source: Author's own elaboration based on ENAHO microdata (2012-2023)

Finally, figure 6 shows the relationship between the objective income decile and the perceived income decile (Panel A), as well as between the objective income decile and the bias (Panel B), broken down by province. This allows us to visualize the relationship between the socioeconomic level of the area (represented by the average objective income decile) and the average bias in the income distribution. In line with the previously obtained results and the relevant literature (Cruces et al., 2013), it is observed that, the higher the purchasing power within the province —understood as the "reference group" individuals use when estimating their position in the distribution—, the higher the perceived decile and the bias.

Figure 6: Perceived Income Decile and Average Bias Relative to the Objective Income Decile by Province



Source: Author's own elaboration based on ENAHO microdata (2012-2023)

3 Related Literature

This section reviews the relevant literature for the development of this paper. The studies analyzed can be grouped into two main lines of research: on the one hand, those examining perceptions—often erroneous—about income distribution; on the other, those investigating the effects of the agrarian reform in Peru.

3.1 Income Perceptions

This paper draws on the literature related to individual perceptions of one's position in the income distribution. In recent years, numerous studies have identified systematic biases in the estimation of individuals' positions in the income scale, both locally and globally. This difficulty in accurately estimating one's real position in the distribution is attributed to the volume of information required, which is not always available. In the absence of some of this information, determining one's position in the income distribution becomes a statistical inference problem. Individuals rely on an imperfect measure, such as their location in the distribution within a "reference group" (usually tied to their place of residence), to infer their position in the national distribution. However, if the individual does not consider factors such as the relative size of their reference group or the selection process that led to its formation, extrapolating these results to the overall population will be flawed.

This phenomenon is significant, as, according to the canonical model of Meltzer and Richard (1981), the degree of inequality in a society determines its levels of redistribution. However, if individual perceptions—and not objective facts—shape political preferences (in this case, preferences for redistribution), the Meltzer-Richard model should be reformulated to consider perceived inequality rather than observed inequality as the relevant variable for determining redistributive preferences and, thus, redistributive intensity. Models incorporating distributive perceptions help explain empirical regularities in the relationship between inequality and redistributive preferences that the traditional model fails to account for (Choi, 2019; Iacono and Ranaldi, 2021; Windsteiger, 2022). This finding is not only theoretically relevant but also significant for public policy, as preferences for redistribution are shaped by individual perception, not by one's position in the income distribution. Thus, individuals may support levels of redistribution that do not align with their own interests if their perception is inaccurate. Correcting these misperceptions could alter the levels of support or opposition to redistribution policies. Numerous studies quantifying this bias have also conducted information provision experiments to assess whether correcting these biases impacts preferences for redistribution, though the results have been inconclusive (Cruces et al., 2013; Karadja et al., 2017; Fehr et al., 2022).

A recurring finding in this literature is the tendency of individuals to perceive themselves as part of the middle deciles of the income distribution, even when they truly belong to the extremes. The explanation for this empirical regularity aligns with the reference group hypothesis mentioned earlier: if the formation process of these groups is not independent of income, high-income individuals tend to be surrounded by others with high incomes, while low-income individuals group with others in similar situations. Using relative position within their reference group as an estimator for their location in the national distribution, high-income individuals tend to underestimate their position, while low-income individuals overestimate it, creating what is known as the "central bias" or "middle-class bias." This bias has been documented in several studies. For example, Cruces et al. (2013) found this bias by combining administrative data on the income distribution at the national level with representative surveys of the population of Greater Buenos Aires. Fehr et al. (2022) found evidence of this phenomenon when asking a representative sample of the German population about their position in the national income distribution. Similar results were reported by Hvidberg et al. (2023) for Denmark and by Gassmann and Timár (2024) in the Kyrgyz Republic. Other studies (Gimpelson and Treisman, 2018; Hoy and Mager, 2021; Bublitz, 2022) support the existence of this phenomenon internationally.

However, the reference group hypothesis is consistent with other possible results. Just as national income distributions are not unique, the shape of the distribution within the reference group is not singular, and different functional forms can lead to variations in the bias. Additionally, changes in the internal distribution of the reference group can modify both the magnitude and direction of the bias. In this regard, middle-class bias is not universally observed in all contexts. For example, Karadja et al. (2017) found that nearly 90% of respondents underestimated their true position in the distribution using a representative sample of the Swedish population. This finding contrasts with the proportionality observed in previous studies between those who underestimate and those who overestimate their position. Furthermore, factors unrelated to the lack of information about the income distribution can also influence individual perceptions. In a recent study, Douenne et al. (2024) demonstrated, using a representative survey of the Dutch population, that individuals do not correctly differentiate between income and wealth, nor do they distinguish accurately between levels of inequality in each of these domains. These results suggest that people perceive a single form of economic inequality, considering both income and wealth as indicators of inequality levels. In the context of this paper, these findings imply that both changes in the distribution of land ownership (as a measure of wealth) resulting from the agrarian reform and variations in the income distribution within districts exposed to the reform could have affected the magnitude of this bias, with consequences that still persist.

3.2 Land Reform

This paper also builds on the literature concerning the effects of the land reform in Peru. The agrarian reform has been a tool of public policy implemented at different historical moments and regions of the world, both under democratic and authoritarian governments. Through the expropriation and redistribution of land, these reforms have been promoted in rural contexts with the aim of ensuring access to land, strengthening property rights, fostering economic development, reducing poverty and inequality, and mitigating social unrest to prevent civil conflicts or political uprisings. However, the empirical evidence has not reached definitive conclusions regarding the success or failure of these reforms, as their impact heavily depends on specific contextual factors (Bhattacharya et al., 2019).

In Peru, the agrarian reform took place between 1969 and 1980. Aimed at reducing inequality in land ownership, the military regime of the time expropriated properties exceeding the 150-hectare threshold. This process resulted in one of the most significant agrarian reforms in Latin America, with the expropriation and redistribution of approximately half of the land. Several studies have analyzed the immediate and long-term effects of this reform.

The study by Albertus and Popescu (2020) suggests that the reform did not promote economic development in the affected districts; on the contrary, those where the reform was more intense experienced economic stagnation and increased poverty. In line with these results, Albertus et al. (2020) shows that the reform had a negative effect on human capital accumulation, as it increased the opportunity cost of education. On the other hand, Albertus (2020) identifies an inverted U-shaped relationship between the intensity of the reform and the occurrence of subsequent civil conflicts. In districts with deep reforms, social unrest decreased, and the likelihood of conflicts was reduced. In contrast, in districts where the reform was not sufficiently strong, the coexistence of winners and losers generated resentment and increased the likelihood of conflict. Finally, Condor (2024) finds that the reform increased the participation of candidates from marginalized ethnic groups in district elections, although this did not translate into greater electoral success.

In summary, these studies demonstrate that the agrarian reform in Peru has had significant and lasting effects on variables such as physical and human capital accumulation, productivity, poverty, and socio-political conflict. Although precise information on its effects on inequality is lacking, the results presented suggest possible changes in individual perceptions.

4 Data and Methodology

To evaluate the causal effect of the agrarian reform on individuals' distributive perceptions, data from two sources are used: the National Household Survey (ENAHO) and a database on the agrarian reform[2].

The ENAHO is a nationally representative survey conducted annually by the National Institute of Statistics and Informatics (INEI) since 1997. Its questionnaires include demographic, labor, and governance-related information, such as transparency and democracy perceptions. To the best of our knowledge, this study is the first to analyze distributive perceptions using the module on opinions about the standard of living[3]. For our analysis, we use ENAHO data

[2]The data are available at this link.

[3]This module is only applied to the head of the household or their partner.

from the period 2012-2023[4] and aggregate the information at the district level to construct a pooled cross-section.

On the other hand, we use the agrarian reform database developed by Albertus (2020), who examined official and historical documents related to the agrarian reform process between 1969 and 1980. This database provides information on the current location of districts that were part of the reform, as well as the distances from the peripheral boundary to the centroid of the agrarian zone.

Additionally, we complement this information with district-level data from alternative sources. To identify the districts that participated in the mita minera, we use the database constructed by Dell (2010), which is included due to the relevance of historical factors that may be correlated with current perceptions, given the long-term effects of the mita on human capital accumulation. Furthermore, we incorporate information about district area (in km^2), altitude, and arable land, characteristics associated with the level of land redistribution[5]. These last two variables were calculated from the United Nations Food and Agriculture Organization (FAO) Global Agro-Ecological Zones database.

Table 1 presents a summary of the variables used in the analysis.

Table 1: Summary statistics

Variable	**Mean**	**Std. Dev.**	**Min.**	**Max.**	**N**
Bias	-2.58	1.42	-7.43	2.71	7096
Formality (Proportion)	0.10	0.07	0	0.4	4057
Perceived Stability (Proportion)	0.33	0.23	0.01	1	6131
Gini Index	0.28	0.1	0.02	0.73	7248
Inside Agrarian Zone Core (=1)	0.55	0.5	0	1	7248
Redistributed Area (%)	17.15	25.7	0	100	7248
Altitude (km a.s.l)	2.32	1.49	0	4.84	7248
Mita Zone (= 1)	0.15	0.36	0	1	7248
Area (hundreds of km^2)	6.95	20.17	0.02	292.35	7248
Cultivable land (% area)	7.78	11.17	0	90	7248

4.1 Empirical Strategy

To examine the impact of the agrarian reform on distributive perceptions, an instrumental variables (IV) strategy is employed, based on the affiliation with the central department of the reform, where exposure to this policy was more intense (Albertus, 2020; Condor, 2024). The reform implemented an institutional strategy designed to limit the reaction capacity of landowners, using agrarian zones as administrative units to accelerate the process. However, the selection of a central department was not random, which introduces an endogeneity problem: if this choice was driven by unobservable characteristics correlated with the outcome of interest, the identification of the causal effect would be compromised.

To address this problem, a binary variable is constructed that classifies districts based on their location inside or outside the central department. Unlike Albertus (2020), who explores

[4]The choice of this period is due to the inclusion of the question on distributive perceptions in the survey.

[5]For example, higher-altitude areas, mainly in the Peruvian highlands, are less likely to have arable land, which affects land redistribution.

marginal variations in the intensity of the reform around the departmental boundary, our approach provides a clearer identification of the level of exposure.[6] The delimitation of agrarian zones was carried out for research and development purposes, without prior knowledge of the topography or spatial distribution of the population. Thus, it does not respond to sociodemographic or economic criteria that could influence perceptions of income distribution. Within an agrarian zone, affiliation with the central department is effectively random. According to the National Institute for Agrarian Innovation (INIA), which holds the original maps of the agrarian reform, the zones were represented by oval-shaped boundaries that slightly varied between versions. Although the initial intent was to group districts with similar ecological characteristics, the territorial boundaries of these zones were defined arbitrarily, without considering departmental boundaries or pre-existing agrarian conditions. Additionally, the delimitation of agrarian zones was carried out before the First Agrarian Census of 1961, suggesting limited knowledge of the land's productive structure at that time. Therefore, it is reasonable to assume that affiliation with the central department within an agrarian zone is exogenous, as it was not influenced by pre-existing socio-economic or developmental factors.

We evaluate the validity of the instrumental variable design based on the two key assumptions: relevance and validity of the instrument. Regarding the relevance assumption, the first stage of the estimation finds a positive and statistically significant correlation at the 1% level between affiliation with the central department and the percentage of land redistributed, which satisfies this assumption.

The validity assumption (or exclusion restriction) cannot be directly tested, but we present historical and institutional arguments supporting its plausibility. A common issue in such designs is that affiliation with the central department, defined by administrative boundaries, could correlate with other departmental policies affecting perceptions of income distribution. However, the historical and institutional context of Peru strengthens the plausibility of affiliation with the central department being a valid instrument. Specifically, the country has historically been a highly centralized state, with political and economic power concentrated in Lima, and departments have primarily operated as administrative units subordinate to the central government, with limited autonomy to design and implement their own policies (Aguilar, 1994; Schonwalder, 2003). This centralization was reflected in the agrarian reform, which was designed and implemented centrally from Lima, without considering regional heterogeneity and following administrative boundaries set for purposes unrelated to the reform (Albertus, 2020).

Throughout Peru's history, multiple attempts at decentralization have failed, maintaining uniformity in the implementation of policies, particularly in key areas like the agrarian reform (Aguilar, 1994). Although the reform was applied nationwide, departments lacked autonomy to design differentiated policies within their territories. Therefore, we can argue that affiliation with the central department satisfies the exclusion restriction, as it is not correlated with the error term or with perceptions of income distribution.

Following Albertus (2020), we restrict the sample based on two criteria: i) districts with a population of fewer than 75,000 inhabitants, and ii) agrarian zones 1 to 7 and 11. The selection of agrarian zones is based on three conditions: a) the agrarian reform was applied in the zone; b) the geographic boundaries remained unchanged, which excludes zone 10, which was split into zones 10 and 13; and c) the zone does not align completely with departmental

[6]Condor (2024) explores variations in the distance to agrarian offices as a measure of exposure.

boundaries, excluding zone 12, which coincides with the boundaries of the department of Puno. Additionally, condition a) excludes zones 8 and 9, located in the Peruvian Amazon, where the reform had no effect.

Once the design is validated and the sample is delineated, we estimate the effect of the agrarian reform on the bias of income distribution perceptions using two-stage least squares (2SLS). Affiliation with the central department of the agrarian zone is used as a source of exogenous variation in exposure to the agrarian reform. The model to be estimated is as follows:

$$Y_{d,z} = \hat{D}_{d,z}\delta + X'_{d,z}\beta + \varepsilon_{d,z} \tag{1}$$

where $Y_{d,z}$ is the variable of interest in district d of agrarian zone z. The vector X includes district-level control variables that are time-invariant (altitude, cultivable area, surface area in km^2) and historical (mining mita zone). $\hat{D}_{d,z}$ represents the predicted exposure to the agrarian reform in the first stage, measured by the percentage of redistributed area in district d of agrarian zone z, obtained from the following first-stage model:

$$D_{d,z} = \gamma_z + Z_{d,z}\pi + X'_{d,z}\theta + \nu_{d,z} \tag{2}$$

where $D_{d,z}$ is the actual percentage of redistributed area in district d of agrarian zone z, and $Z_{d,z}$ is the instrumental variable, defined as belonging to the central department within the agrarian zone.

5 Results

5.1 Main Results

Using an IV approach with a 2SLS model, we find that a 1 percentage point increase in the redistributed area reduces the bias by 0.127 points on average, controlling for time-invariant district characteristics such as altitude, cultivable area, total surface area, and mita zone. The results are statistically significant at the 1% level. In particular, in districts belonging to the central department of the agrarian zone, the absolute magnitude of the bias is higher compared to peripheral districts, indicating a more pronounced underestimation of relative position in the income distribution. This suggests that the massive land redistribution may have generated more distorted perceptions of inequality. Table 2 presents the estimation results.

Table 2: Effect of Agrarian Reform on the Bias in Distribution Perceptions

	Bias	
	(1)	(2)
Redistributed Area (%)	-0.127***	-0.163***
	(0.035)	(0.041)
Controls	Si	No
Mean Var. Dep.	-2.58	-2.59
Observations	7096	7096

Notes: The table reports 2SLS estimates of the effect of land redistribution on the bias in income distribution perceptions. The redistributed area (%) is instrumented using the binary variable indicating belonging to the central department of the agrarian zone. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, ***

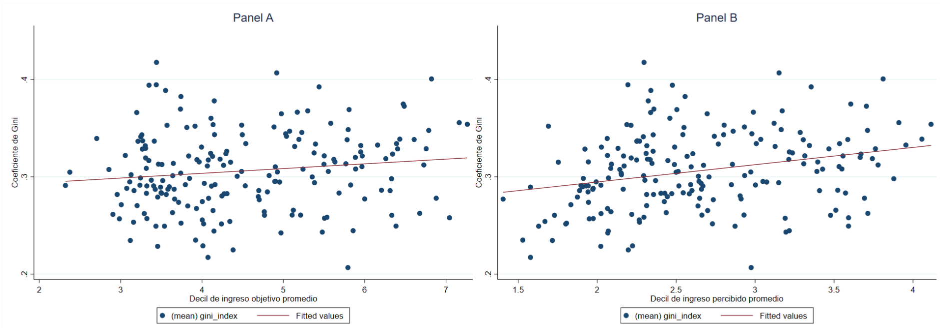
5.2 Mechanisms

The results show that the absolute bias increases in districts with greater exposure to the agrarian reform. We identify three possible transmission mechanisms that could explain this effect: inequality, perceived stability, and labor formality.

Historically, agrarian reform emerged as a response to the landlords' abuses of power over workers, which generated political instability and fostered the rise of subversive movements. In this context, land redistribution aimed to reduce inequality and mitigate potential uprisings.

Testing the inequality mechanism is challenging, especially in the long run. Ideally, inequality measures before, during, and after the reform would be required, but data availability is limited. Given this constraint, we assess the long-term impact of the reform on inequality using the Gini coefficient for the period relevant to this study. Figure 7 shows the relationship between the provincial-level Gini coefficient and the average target income decile (Panel A) and the average perceived income decile (Panel B). A positive relationship with the Gini coefficient is observed, where the values of the average perceived decile are bounded between 2 and 4, while the target decile ranges from 2 to 8, highlighting the presence of a negative bias at the provincial level.

Figure 7: Relationship between the Gini Coefficient and Objective and Perceived Income Decile by Province



Source: Author's own elaboration based on ENAHO microdata (2012-2023)

Table 3 presents the 2SLS estimation results, using the Gini coefficient as the dependent variable to analyze its role as a mechanism. We find that a one-percentage-point increase in redistributed land has a negative and significant effect at the 5% level on inequality. In other words, in the last decade, living in an area with greater agrarian reform intensity is associated with lower inequality.

Since inequality acts as a mechanism influencing bias, this suggests that lower inequality is linked to a greater negative bias. However, theoretically, a reduction in income disparity could decrease the underestimation of economic position or even lead to an overestimation in contexts where the socioeconomic environment was below average. Nevertheless, the literature on agrarian reform indicates that it had no significant impact on economic development or human capital accumulation in the affected regions (Albertus et al., 2020; Albertus and Popescu, 2020). This suggests that its effects were not sustainable over time. In this context, the negative bias can be explained by the fact that inhabitants of these areas tend to perceive themselves as poorer compared to other regions. Thus, although income inequality decreased, the perception of relative position remains more pessimistic.

Table 3: Inequality

	Gini Index	
	(1)	(2)
Redistributed Area (%)	-0.003**	-0.002*
	(0.001)	(0.001)
Controls	Si	No
Mean Var. Dep.	0.278	0.278
Observations	7248	7248

Notes: The table reports 2SLS estimates of the effect of land redistribution on the bias in income distribution perceptions. The redistributed area (%) is instrumented using the binary variable indicating belonging to the central department of the agrarian zone. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** Additionally, we explore two other mechanisms linked to living conditions and the labor environment: perceived stability and formality. To measure perceived stability, we use data from the governance module of ENAHO[7].

The model results indicate that a one-percentage-point increase in redistributed land leads, on average, to lower perceived stability in areas more exposed to the reform, with statistical significance at the 1% level. The relationship with bias is direct: a lower sense of stability causes individuals to perceive themselves as poorer than they actually are, reinforcing the underestimation of their economic position.

The third identified mechanism is related to the proportion of formal employees in areas affected by agrarian reform. We find that a one-percentage-point increase in redistributed land is associated with a rise in the proportion of formal workers. This result is intuitive since land redistribution during the reform was implemented through the creation of cooperatives, which led to the formalization of businesses in these regions. Although the relationship with bias is not direct, it can be explained by the fact that the Peruvian economy is highly centralized and bureaucratic. Despite the reform fostering greater prosperity, the cooperatives created were supervised by government-appointed bureaucrats, limiting the direct control of farmers over the management and administration of these entities.

Table 4: Perceived Stability and Formality

	Perceived Stability (Proportion)		Formality (Proportion)	
	(1)	(2)	(3)	(4)
Redistributed Area (%)	-0.019***	-0.023***	0.011**	0.013***
	(0.005)	(0.005)	(0.006)	(0.005)
Controls	Si	No	Si	No
Mean Var. Dep.	0.33	0.33	0.10	0.10
Observations	6131	6131	4057	4057

Notes: The table reports 2SLS estimates of the effect of land redistribution on the bias in income distribution perceptions. The redistributed area (%) is instrumented using the binary variable indicating belonging to the central department of the agrarian zone. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, ***

5.3 Heterogeneous Effects

The main results cover the national territory, excluding agrarian zones based on the previously mentioned criteria. To explore potential heterogeneous effects, we estimate the instrumental variables model for specific groups of individuals.

Table 5 presents the results by cohorts. We analyze whether the effects vary by age group, as one would expect that those who lived through the implementation of the agrarian reform, and were therefore more exposed, exhibit differences in the observed outcomes.

[7]We used a question where respondents are asked whether the household perceives its income as stable, somewhat stable, or very unstable. Stable household were coded as one, zero otherwise.

Table 5: Cohorts

	Cohort (1950s)		Cohort (1960s)		Cohort (1970s)		Cohort (1980s)		Cohort (1990s)		Cohort (2000s)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Redistributed Area (%)	-0.158***	-0.196***	-0.117***	-0.160***	-0.089***	-0.134***	-0.133***	-0.165***	-0.138	-0.172	-0.487	-0.121
	(0.045)	(0.049)	(0.031)	(0.041)	(0.032)	(0.043)	(0.043)	(0.053)	(0.088)	(0.112)	(2.273)	(0.190)
Controls	Si	No	Si	No	Si	No	Si	No	Si	No	Si	No
Mean Var. Dep.	-3.01	-3.01	-2.73	-2.73	-2.09	-2.09	-1.84	-1.84	-2.44	-2.44	-3.32	-3.32
Observations	6204	6204	6337	6337	6157	6157	5128	5128	2510	2510	193	193

Notes: The table reports 2SLS estimates of the effect of land redistribution on the bias in income distribution perceptions. The redistributed area (%) is instrumented using the binary variable indicating belonging to the central department of the agrarian zone. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, ***

To compute the cohorts, we use ENAHO to identify the year of birth and assign a cohort accordingly. We collapse the data by year, district, and cohort. We observe that in districts where the intensity was higher, i.e., the central departments, individuals born between the 1950s and 1980s exhibit an increase in absolute bias. The results are significant at the 1% level.

This effect is consistent with the general results in Table 2. However, for younger groups, the estimates are not statistically different from zero. This can be explained by the implementation period of the reform, which spanned from 1969 to 1980. Therefore, those born in the latter decade may have been influenced by the changes resulting from land redistribution. In contrast, individuals born from the 1990s onward grew up in a context where the reform was already consolidated, implying that they did not experience a direct impact on their perceptions of income distribution.

Finally, we perform a heterogeneous effects analysis for identifiable reference groups. We compute reference groups using a question that identifies with whom or what group the household head identifies. Cruces et al. (2013) found a strong relationship between the socioeconomic level of the place of residence and bias perception. Table 6 presents the 2SLS estimation results using reference groups. We show that a one percentage point increase in the redistributed area leads to an increase in absolute bias for those who consider their reference group to be geographic, their peasant community, or their religious group. All results are statistically significant at the 1% rejection level, except for ethnic or racial groups.

Table 6: Resultados IV - Grupo de referencia

	Geographic		Ethnicity and Race		Peasant communities		Religious group	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Redistributed Area (%)	-0.145***	-0.194***	-0.527	-0.989	-0.081**	-0.097***	-0.162***	-0.231***
	(0.048)	(0.063)	(0.671)	(2.006)	(0.032)	(0.032)	(0.050)	(0.079)
Controls	Si	No	Si	No	Si	No	Si	No
Mean Var. Dep.	-2.70	-2.70	-2.85	-2.85	-2.48	-2.48	-2.78	-2.78
Observations	5812	5812	1878	1878	4423	4423	4600	4600

Notes: The table reports 2SLS estimates of the effect of land redistribution on the bias in income distribution perceptions. The redistributed area (%) is instrumented using the binary variable indicating belonging to the central department of the agrarian zone. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, ***

6 Conclusions

This study analyzes the long-term effects of the Peruvian agrarian reform, a nationwide redistributive policy, on individuals' perceptions of income distribution. Using a previously underexplored dataset on land reform and leveraging the governance, democracy, and transparency module of ENAHO, this paper provides novel evidence on the implications of land redistribution over time.

The results suggest that the agrarian reform widened the gap between perception and reality, leading to a greater underestimation of relative economic position in districts with higher levels of intervention.

Furthermore, we identify several transmission mechanisms. First, the Gini coefficient shows that areas exposed to the reform exhibit lower inequality, though with a relatively poorer and more homogeneous population. Second, greater exposure to the reform is associated with a lower proportion of individuals who consider their economic situation stable. Finally, the analysis of the proportion of formal workers reveals an increase linked to the creation of cooperatives during the reform.

The heterogeneous effects analysis reveals that the temporal proximity to the reform influences the magnitude of the perception-reality gap. In the most affected districts, individuals born and living near the implementation period exhibit a greater absolute bias compared to more recent cohorts (born after 1990).

This study has two main limitations stemming from the nature of the data. First, since ENAHO is a representative survey, it does not uniformly cover the entire national territory, preventing the calculation of the bias measure for certain districts and years. Second, the lack of pre-reform data limits the ability to control for initial socio-economic factors that could improve the accuracy of the estimates.

In conclusion, the agrarian reform implemented during Velasco's government played a key role in shaping perceptions of income distribution, especially among those who directly experienced its effects. These findings contribute to the literature on redistribution and distributive perceptions, offering a historical perspective on the persistent effects of redistributive policies on individuals' perceptions of inequality.

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